

# **Bodax SACCO Management System Project Proposal.**

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## **1. Project Title.**

Bodax SACCO Management System for Boda Boda Riders in Mbarara, Uganda.

## **2. Executive Summary.**

Bodax SACCO Management System is a digital savings, loans, withdrawals, and reporting platform designed for boda boda SACCO groups, beginning with the Mile 4 Boda Boda Stage in Mbarara. The project replaces manual notebook-based SACCO record keeping with a secure web system that allows members to view their savings, request loans, track repayments, and receive clear financial updates.

The system is built as a full-stack application using React for the user interface, Node.js and Express for the API, and PostgreSQL for the SACCO database. It supports three main roles: Member, Treasurer, and Chairman. Each role has a different dashboard and permission level so that financial operations are controlled, traceable, and transparent.

Field testing at Mile 4 Stage showed that riders value instant visibility of savings balances, statement history, and loan status. The same test also identified improvements required before wider rollout, including phone-number login, confirmation prompts before financial actions, SMS-style notifications, better mobile layouts, and clearer local-language-friendly wording.

## **3. Background.**

Many boda boda SACCO groups still manage savings and loans using paper books, receipts, and verbal updates from treasurers. This creates delays, errors, mistrust, and difficulty producing reports for leaders. Members often need to physically contact the treasurer to confirm their balance or loan status.

For a busy stage such as Mile 4 in Mbarara, this approach becomes difficult as membership grows. Riders need a simple way to confirm deposits, see loan eligibility, follow repayment obligations, and trust that their records are accurate. SACCO leaders also need faster reporting on active members, total savings, loan arrears, top savers, defaulters, and withdrawals.

Bodax SACCO addresses these needs through a role-based digital platform that is simple enough for ordinary riders while giving the treasurer and chairman stronger control over financial workflows.

#### **4. Problem Statement.**

The current manual SACCO process has the following challenges:

- Savings and loan records are vulnerable to human error.
- Members cannot instantly confirm whether their deposit was recorded.
- Similar member names can lead to wrong savings entries.
- Loan requests and approvals take time because they depend on manual communication.
- Members lack automatic reminders for due or overdue loans.
- Chairman and treasurer reports are slow to compile.
- Paper records make it hard to detect defaulters, arrears, or collection trends quickly.
- Riders are more comfortable with phone numbers and SMS than email-based systems.
- Small-screen Android phones need a mobile-first experience.

#### **5. Proposed Solution.**

The proposed solution is to complete and strengthen Bodax SACCO Management System into a field-ready digital SACCO platform. The system will provide:

- Secure login and role-based access for Members, Treasurers, and Chairmen.
- Member registration with member number, phone number, stage, next of kin, and account status.
- Savings recording and member savings summaries.
- Loan eligibility checks based on confirmed savings.
- Loan request, approval, issuance, repayment, and overdue tracking.
- Withdrawal request and approval workflows.
- Member statements showing savings, loan repayments, and withdrawals.
- Treasurer and chairman dashboards with operational and governance metrics.
- Reports for top savers, defaulters, collections, income, and expenditure.
- Mobile usability improvements for low-cost Android devices.
- Notifications and transaction confirmation features to improve trust.

#### **6. Project Objectives.**

## 6.1 Main Objective.

To develop a reliable, secure, and user-friendly SACCO management system that improves transparency, financial accountability, and service delivery for boda boda SACCO members and leaders.

## 6.2 Specific Objectives.

- Digitize member registration and SACCO profile management.
- Enable accurate recording of savings deposits by the treasurer.
- Allow members to view savings balances, recent transactions, and statements.
- Automate loan eligibility checks using the SACCO savings multiplier rule.
- Support loan requests, approval decisions, loan issuance, and repayment tracking.
- Track withdrawals through request and approval workflows.
- Provide dashboards and reports for treasurer operations and chairman oversight.
- Improve field usability through mobile-first design, clear language, and phone-based access.
- Reduce financial data-entry mistakes using confirmation screens and validation improvements.

## 7. Scope.

### 7.1 In Scope.

- Web-based SACCO application for members, treasurer, and chairman.
- Authentication using JWT and role-based authorization.
- PostgreSQL database for users, roles, members, savings, loans, repayments, withdrawals, and reports.
- Member dashboard, treasurer dashboard, and chairman dashboard.
- Savings, loan, withdrawal, statement, and reporting modules.
- Mobile responsiveness improvements.
- Field-test-driven improvements for login, confirmations, amount inputs, and notifications.
- Deployment configuration for hosted frontend and backend environments.

### 7.2 Out of Scope for Initial Completion.

- Full mobile app in Android native code.
- Integration with live mobile money payment collections.
- Advanced accounting ledger integration.

- Biometric authentication.
- Full offline-first synchronization for all modules, unless time allows after core features.
- Multi-branch SACCO support beyond the initial Mbarara pilot.

## 8. Stakeholders.

| Stakeholder            | Role in the Project | Main Interest                                                                     |
|------------------------|---------------------|-----------------------------------------------------------------------------------|
| SACCO Members          | Primary end users   | View savings, request loans, receive confirmations, track repayments              |
| Treasurer              | Operational user    | Register members, record savings, issue loans, record repayments, review requests |
| Chairman               | Oversight user      | Monitor performance, arrears, defaulters, top savers, and financial health        |
| Bodax Development Team | Implementation team | Build, test, deploy, and maintain the system                                      |
| SACCO Committee        | Governance body     | Approve policies, rollout decisions, and operating procedures                     |
| Field Test Riders      | Pilot users         | Provide usability feedback from real stage conditions                             |

## 9. Functional Requirements.

### 9.1 Member Management

- Register SACCO members with member number, full name, phone number, stage, national ID, next of kin, and status.
  - Search and view members by name, phone number, or member number.
  - Update member records and activate or deactivate members.
  - Assign member login credentials.

## **9.2 Authentication and Authorization.**

- Allow users to log in using credentials.
  - Protect all SACCO data behind authentication.
  - Restrict system actions according to role:
- Members view their own dashboard, loans, statements, and profile.
- Treasurers manage members, savings, loans, repayments, and approvals.
- Chairmen view oversight dashboards, analytics, and reports.

## **9.3 Savings Management.**

- Record member savings deposits.
  - Store transaction date, amount, notes, confirmation status, and treasurer identity.
  - Summarize savings by week, month, year, and total.
  - Show recent savings and statements to members.

## **9.4 Loan Management.**

- Check loan eligibility using the rule that a member can borrow up to 3 times confirmed savings.
  - Prevent loans for inactive members.
  - Prevent new loans where a member has an active, overdue, or pending loan request.
  - Allow members to request loans.
  - Allow treasurers to approve or reject loan requests.
  - Calculate interest, total payable, installments, and remaining balance.
  - Track repayments and mark loans as active, completed, or overdue.

## **9.6 Reports and Analytics.**

- Treasurer dashboard: active members, daily collections, weekly collections, monthly collections, active loans, and pending loan requests.
  - Chairman dashboard: total members, total savings, active loans, outstanding loan balance, loan arrears, weekly collections, and monthly collections.
  - Analytics: top savers, defaulters, monthly collection trends, income summary, and expenditure summary.

## **10. Non-Functional Requirements.**

- Security: JWT authentication, password hashing, protected routes, role-based authorization, rate limiting, and HTTP security headers.
- Reliability: database-backed records with transaction handling for loan and withdrawal approvals.
- Usability: simple language, mobile-first layouts, large touch targets, and clear confirmations.
- Performance: indexed database fields for common member, savings, loan, and request queries.
- Maintainability: modular frontend pages and backend routes, controllers, services, validators, and database schema.
- Deployability: environment-based configuration for API, frontend URL, JWT secret, and database URL.

## **11. Technical Approach.**

The system will use the existing full-stack architecture:

- Frontend: React, Vite, React Router, Context API, Axios, Recharts, and Lucide React.
- Backend: Node.js, Express.js, Zod validation, JWT, bcryptjs, Helmet, CORS, Morgan, Express Rate Limit, and PostgreSQL client.
- Database: PostgreSQL, currently prepared for Neon PostgreSQL using tables for roles, users, members, savings, loans, repayments, loan requests, withdrawal requests, withdrawals, and reports.
- Deployment: Vercel or Render-style hosting configuration for frontend and backend services.

## **12. Expected Benefits.**

- Better financial transparency for riders.
- Faster savings recording and statement access.
- Reduced dependency on paper books.
- Improved loan decision-making using savings-based eligibility.
- Faster identification of defaulters and overdue loans.
- Stronger trust between members, treasurer, and chairman.
- Easier reporting for SACCO governance meetings.
- Better preparation for future integrations such as SMS, WhatsApp, OTP, and mobile money.

## **13. Key Risks and Mitigation**

| <b>Risk</b>                                      | <b>Impact</b> | <b>Mitigation</b>                                                                 |
|--------------------------------------------------|---------------|-----------------------------------------------------------------------------------|
| Treasurer records savings under the wrong member | High          | Add confirmation modal showing member name, member number, and amount before save |
| Riders struggle with email login                 | High          | Support phone-number login and prepare OTP integration                            |
| Users enter UGX values with commas               | Medium        | Normalize comma-formatted amounts before validation                               |
| Mobile tables are hard to read                   | Medium        | Convert narrow-screen tables into responsive transaction cards                    |
| Poor network at stages                           | Medium        | Add graceful loading states, retry handling, and plan offline support             |
| Members do not trust digital records             | High          | Show member full name, member number, latest deposit confirmation, and receipts   |
| Loan decisions are not communicated              | Medium        | Add SMS or WhatsApp notification roadmap and in-app request status updates        |

## 14. Implementation Duration.

The proposed completion period is 6 weeks. The work will begin with documentation and critical field-test fixes, then move into workflow completion, reporting, testing, deployment, and pilot preparation.

## Work Breakdown by Week

### Week 1: Documentation Baseline and Critical Safety

Main goal: Establish clear documentation and fix the highest-risk financial workflow issue.

#### Planned activities:

- Prepare project proposal, project documentation, and project plan.
- Review existing modules and confirm system scope.

- Document current business rules for savings, loans, withdrawals, and reports.
- Add confirmation prompts before important financial submissions:
- Record savings.
- Approve loan request.
- Reject loan request.
- Issue loan.
- Record repayment.
- Approve withdrawal.
- Reject withdrawal.
- Ensure confirmation text shows member name, member number, amount, and action.
- Review current validation messages and make them clearer for field users.
- Create a testing checklist for critical financial workflows.

#### Deliverables:

- Project proposal document.
- Project documentation document.
- Project plan document.
- Confirmation workflow design.
- Updated financial action screens.
- Week 1 test notes.

#### Acceptance criteria:

- Documentation exists and reflects the current system.
- Treasurer cannot accidentally submit major financial actions without seeing a confirmation message.
- Confirmation messages use human-readable member details, not only IDs.

## Week 2: Authentication and Member Trust.

Main goal: Make login and member identity easier and more trustworthy for riders.

#### Planned activities:

- Improve login screen wording so riders understand they can use phone number or email where supported.
- Prepare phone-number-first login flow.



- Review member signup and member credential assignment.
- Display member full name and member number prominently on the member dashboard.
- Add or prepare space for member photo/avatar if required.
- Improve profile page to clearly show SACCO identity details.
- Make password change flow clearer.
- Document recommended OTP integration approach for future SMS login.
- Review authentication and authorization edge cases.

#### **Deliverables:**

- Improved login and member identity screens.
- Updated member dashboard identity header.
- Authentication notes in documentation.
- OTP integration recommendation.

#### **Acceptance criteria:**

- A member can clearly identify that the dashboard belongs to them.
- The login experience is less email-dependent in wording and flow.
- Profile information is understandable to ordinary SACCO members.

### **Week 3: Savings, Statements, and Usability.**

Main goal: Improve day-to-day SACCO transactions and small-screen use.

#### **Planned activities:**

- Update UGX amount inputs to accept values such as `50000` and `50,000`.
- Normalize formatted amounts before API submission or validation.
- Improve validation messages for amount fields.
- Improve savings record form layout and field labels.
- Add deposit success details that can act as a simple receipt summary.
- Improve member statements for mobile screens by using responsive cards or better table behavior.
- Improve withdrawal request and withdrawal review screens.
- Confirm statement records show savings, loan repayments, and withdrawals clearly.
- Test on narrow mobile widths representing low-cost Android phones.

#### **Deliverables:**

- Improved amount input handling.
- Mobile-friendly statements view.
- Clear savings success/receipt summary.
- Improved withdrawal workflow screens.
- Week 3 mobile test notes.

#### **Acceptance criteria:**

- Users can enter `50,000` without validation confusion.
- Statements are readable on small screens.
- Savings and withdrawal workflows use clearer member-facing language.

#### **Week 4: Loans, Repayments, Reminders, and Chairman Visibility.**

Main goal: Strengthen the SACCO loan workflow and oversight experience.

#### **Planned activities:**

- Review loan eligibility messages and simplify loan terminology.
- Replace confusing terms where needed:
- Installment count becomes Number of payments.
- Principal becomes Amount borrowed where user-facing.
- Improve loan request status display for members.
- Improve loan approval/rejection screens for treasurer.
- Ensure active, completed, and overdue loans display clearly.
- Improve repayment recording workflow.
- Review overdue loan status refresh behavior.
- Improve chairman view of defaulters and overdue loans.
- Ensure chairman and treasurer views show member names and member numbers instead of technical IDs.
- Prepare SMS or WhatsApp notification event list:
- Deposit recorded.
- Loan approved.
- Loan rejected.
- Loan due soon.

- Loan overdue.
- Withdrawal approved.

**Deliverables:**

- Clearer loan forms and loan status pages.
- Improved overdue loan and defaulter visibility.
- Notification event plan.
- Week 4 loan workflow test notes.

**Acceptance criteria:**

- Members understand loan request fields and status.
- Treasurer can approve/reject and record repayments confidently.
- Chairman can identify overdue loans by member name and amount.

**Week 5: Reporting, Testing, Deployment, and Training Materials.**

Main goal: Stabilize the system and prepare it for a controlled pilot.

**Planned activities:**

- Verify treasurer dashboard totals.
- Verify chairman dashboard totals.
- Verify analytics for top savers, defaulters, collections, income, and expenditure.
- Add or document report interpretation notes for SACCO leaders.
- Test all major API endpoints.
- Test role-based access for Member, Treasurer, and Chairman.
- Review deployment environment variables.
- Confirm health check endpoint works.
- Review Render and Vercel deployment setup.
- Prepare training materials:
  - Treasurer quick guide.
  - Member quick guide.
  - Chairman quick guide.
  - Pilot support checklist.

Deliverables:

- Tested dashboards and reports.
- Deployment checklist.
- User training guides.
- Role-based testing checklist.
- Week 5 test report.

**Acceptance criteria:**

- Dashboard numbers match expected database records.
- Each role can access only the correct screens.
- Deployment steps and environment variables are documented.
- Training guides are ready for SACCO users.

**Week 6: User Acceptance Testing, Final Fixes, and Handover.**

Main goal: Complete final validation and prepare for SACCO committee sign-off.

**Planned activities:**

- Conduct full end-to-end testing:
  - Register member.
  - Assign login.
  - Record savings.
  - View member dashboard.
  - Request loan.
  - Approve loan.
  - Record repayment.
  - Track overdue/completed status.
  - Request withdrawal.
  - Approve withdrawal.
  - View reports.
- Conduct user acceptance testing with sample users.
- Fix defects discovered during final testing.
- Finalize project proposal, documentation, and plan.
- Prepare final handover package.

- Prepare pilot rollout recommendations.
- Record known limitations and future enhancements.

#### **Deliverables:**

- Final tested application.
- Final project documentation.
- Final user guides.
- UAT report.
- Handover notes.
- Pilot rollout recommendation.

#### **Acceptance criteria:**

- Core SACCO workflows run successfully end to end.
- Critical field-test issues are fixed or clearly documented for the next phase.
- SACCO committee can review the project with clear documents and test evidence.

## **15. Success Criteria.**

The project will be considered successful when:

- Members can log in, view balances, request loans, and view statements on mobile phones.
- Treasurers can register members, record savings, review loan requests, issue loans, record repayments, and review withdrawals.
- Chairmen can view analytics, reports, defaulters, and SACCO financial summaries.
- Critical financial actions require confirmation before submission.
- UGX amount fields accept common user formats such as `50000` and `50,000`.
- The application works clearly on small Android screens.
- The system is deployed and ready for a controlled SACCO pilot.
- Project documentation, user guidance, and handover notes are complete.

## **16. Conclusion.**

Bodax SACCO Management System is a practical and relevant solution for boda boda SACCO operations in Mbarara. It already has a strong technical foundation and matches the core workflows of savings, loans, withdrawals, and governance reporting. With the proposed six-week completion plan, the system can move from prototype status to a more field-ready platform that improves transparency, accuracy, and member confidence.